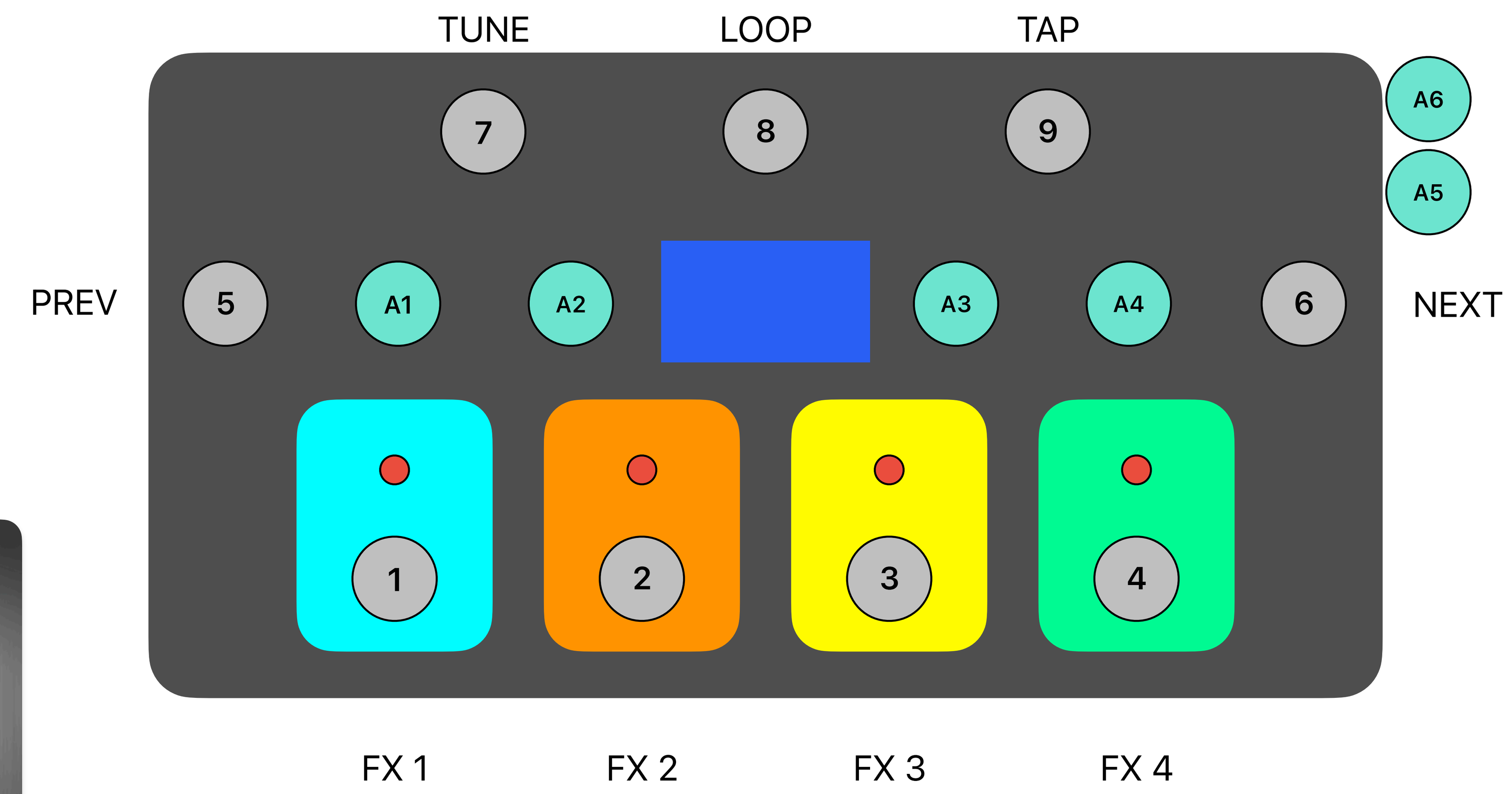

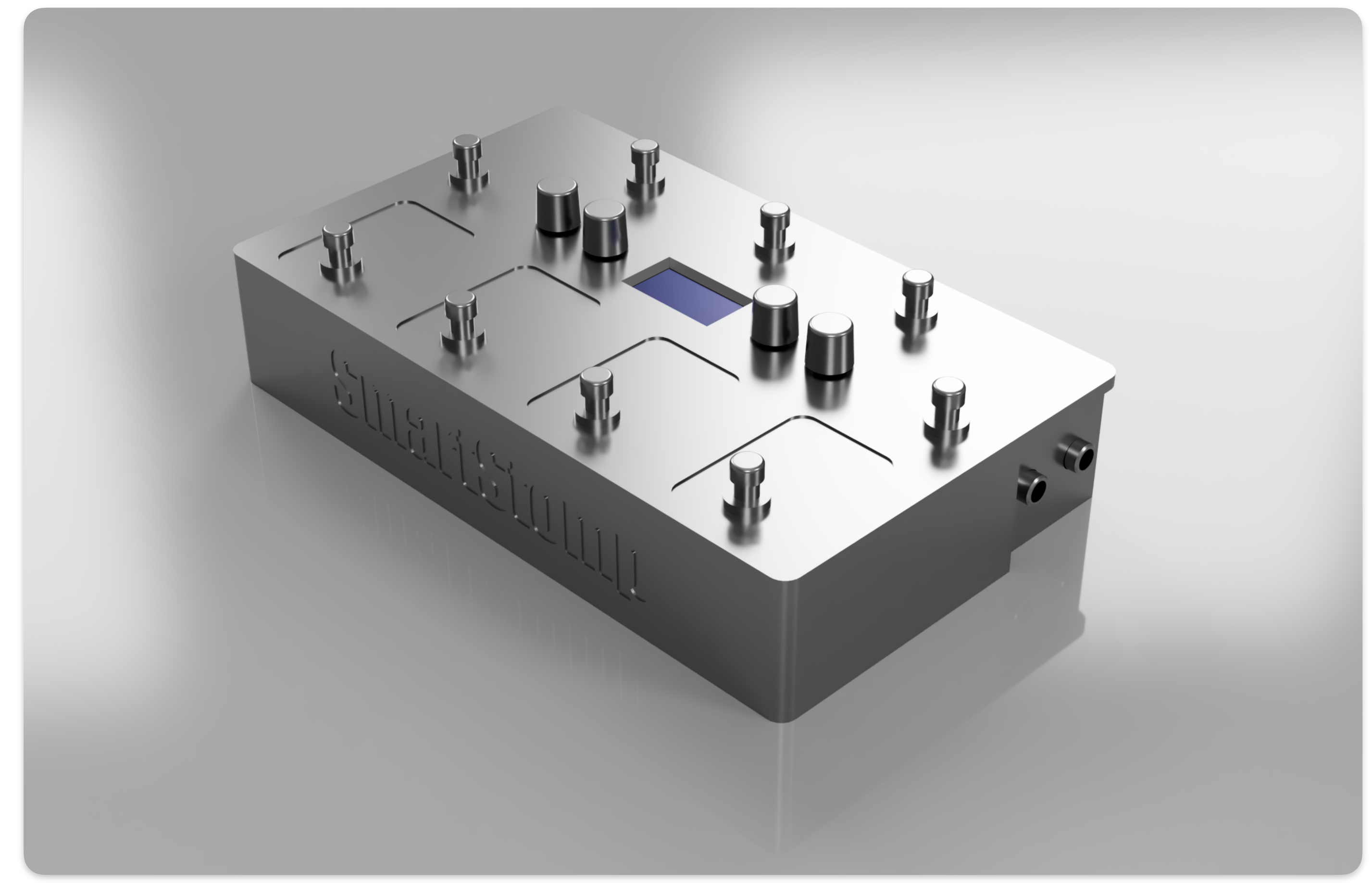

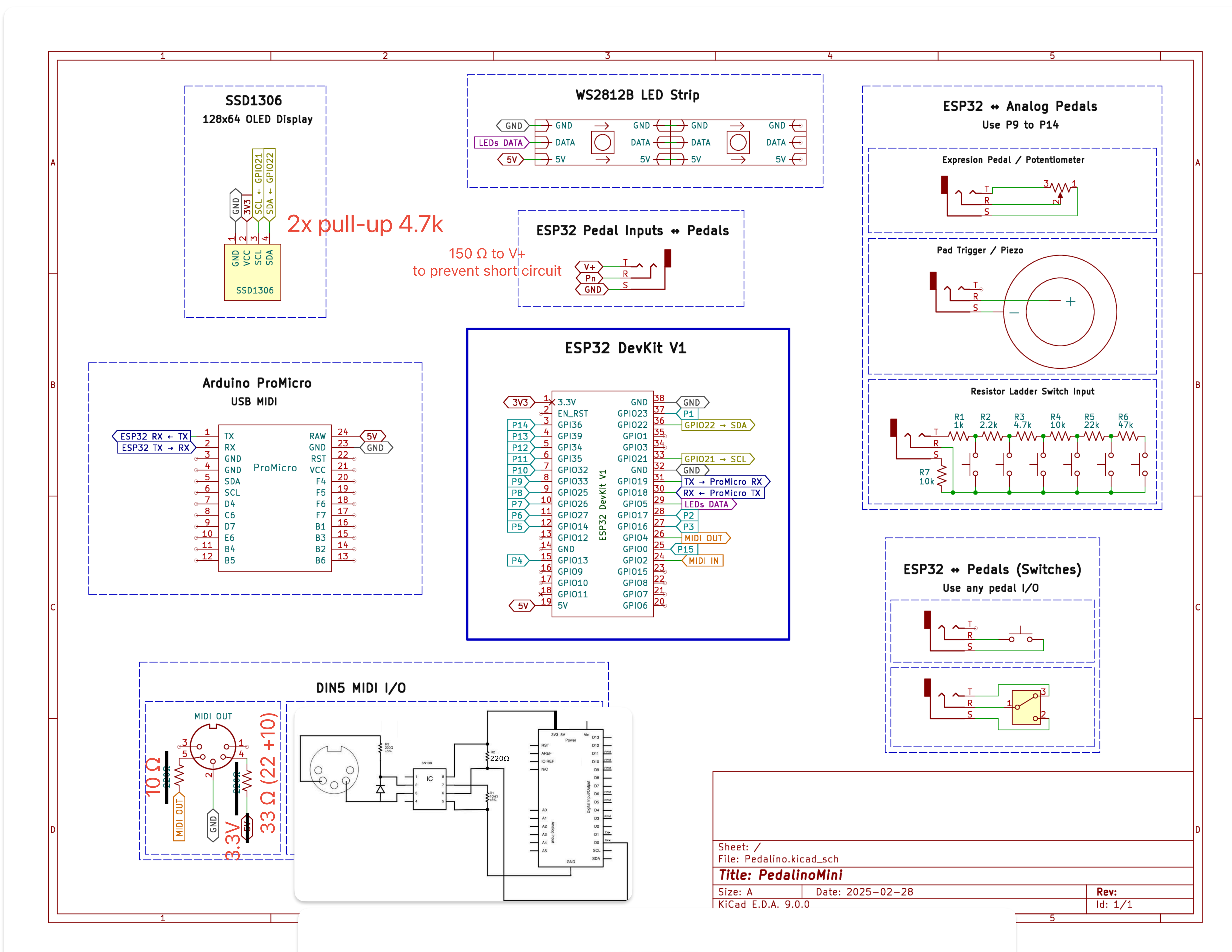

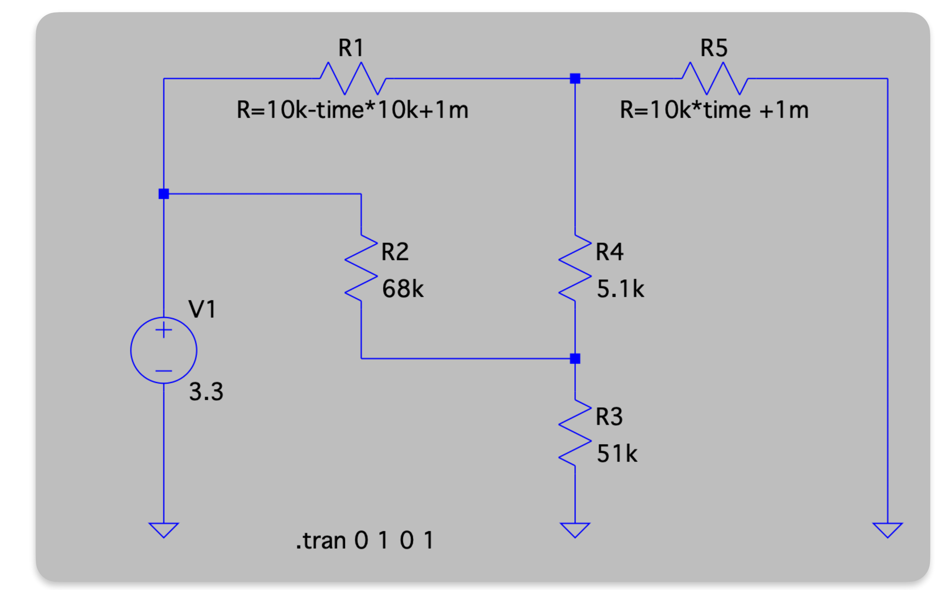

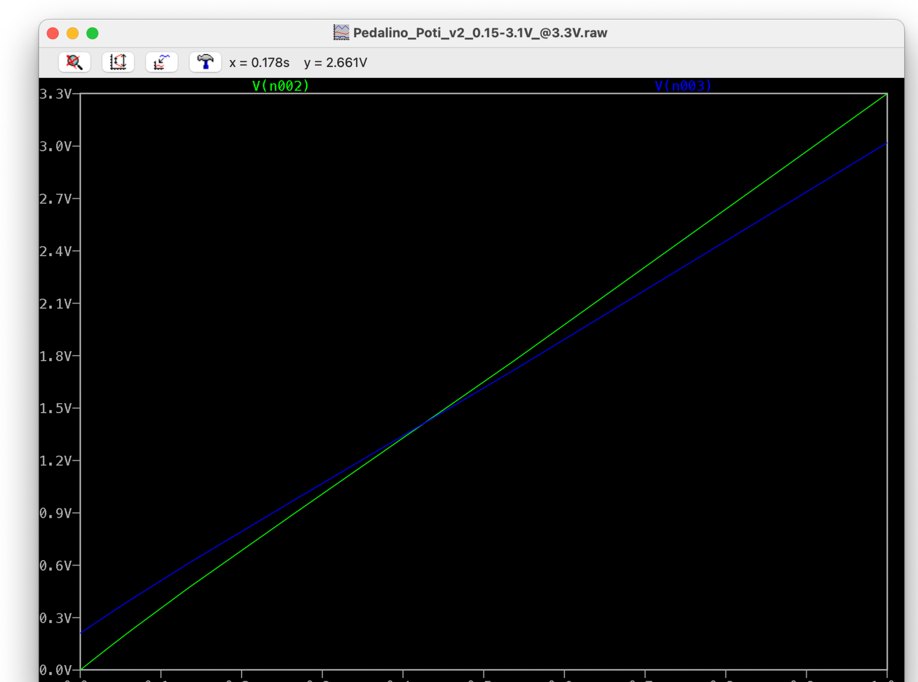

Boot Mode Selection

This guide explains how to select the boot mode correctly and describes the boot log messages of ESP32.

Warning

The ESP32 has a 45k ohm internal pull-up/pull-down resistor at GPIO0 (and other pins). If you want to connect a switch button to enter the boot mode, this has to be a strong pull-down. For example a 10k resistor to GND.

Information about ESP32 strapping pins can also be found in the ESP32 Datasheet, section "Strapping Pins".

On many development boards with built-in USB/Serial, `esptool.py` can automatically reset the board into bootloader mode. For other configurations or custom hardware, you will need to check the orientation of some "strapping pins" to get the correct boot mode:

Select Bootloader Mode

GPIO0

The ESP32 will enter the serial bootloader when GPIO0 is held low on reset. Otherwise it will run the program in flash.

GPIO0 Input	Mode
Low/GND	ROM serial bootloader for esptool
High/VCC	Normal execution mode

GPIO0 has an internal pullup resistor, so if it is left unconnected then it will pull high.

Many boards use a button marked "Flash" (or "BOOT" on some Espressif development boards) that pulls GPIO0 low when pressed.